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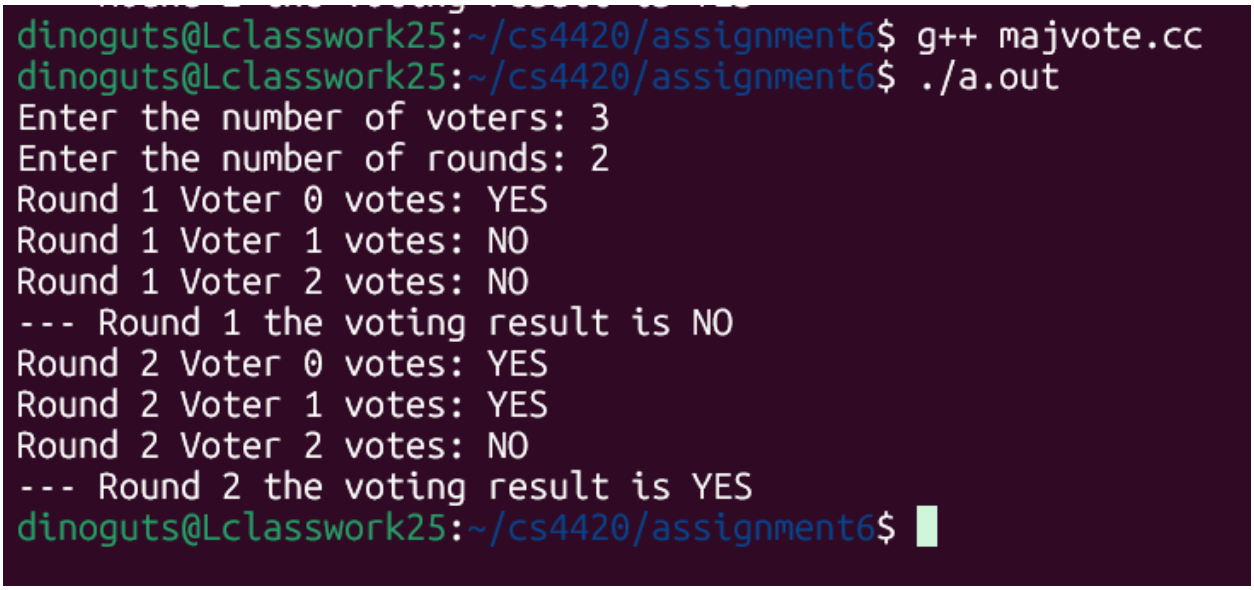
Fall 2026

CS 4420

Assignment 6

The program is named majvote.cc and is compiled using “g++ majvote.cc”. It is run with “./a.out”. When run, the program will keep prompting the user to enter the number of voters until they enter a number greater than or equal to 3, the minimum number of voters. The program will then keep prompting the user to enter the number of rounds until they enter a valid number of rounds, any number greater than or equal to 1. The program creates a number of pipes equal to the number of voters, forks to create voter IDs for every child process, and the conducts rounds of voting there the parent process (the judge) counts votes the voters made and prints them out, as well as the overall vote for the round, and the child processes (the voters) generate random numbers to decide their votes for each round. After all the voting rounds are finished, each pipe is closed.

Screenshot 1: normal operation with an odd number of voters (ties are not possible)



Screenshot 2: normal operation with an even number of voters (ties are possible)

```
dinoguts@Lclasswork25:~/cs4420/assignment6$ g++ majvote.cc
dinoguts@Lclasswork25:~/cs4420/assignment6$ ./a.out
Enter the number of voters: 4
Enter the number of rounds: 3
Round 1 Voter 0 votes: NO
Round 1 Voter 1 votes: NO
Round 1 Voter 2 votes: YES
Round 1 Voter 3 votes: YES
--- Round 1 the voting result is TIE
Round 2 Voter 0 votes: NO
Round 2 Voter 1 votes: NO
Round 2 Voter 2 votes: NO
Round 2 Voter 3 votes: NO
--- Round 2 the voting result is NO
Round 3 Voter 0 votes: NO
Round 3 Voter 1 votes: YES
Round 3 Voter 2 votes: NO
Round 3 Voter 3 votes: YES
--- Round 3 the voting result is TIE
dinoguts@Lclasswork25:~/cs4420/assignment6$
```

Screenshot 3: How processes communicate in this program using pipes

